DAEN 328 Assignment 5: Load and Serve (CSV to PostgreSQL to Streamlit)

Due: Friday, October 30, 11:59 pm on Canvas. 100 points.

Files you need: `Assignment05_Guide.ipynb`, `practice_orders.csv`, `practice_app.py`, `Assignment05_Starter.ipynb`, your Assignment 3 `Cleaned_data.csv`, and your Assignment 4 `EmployeeDB` with its 7 empty tables.

What this assignment is

This is the full path a data engineer builds: take a cleaned file, **load** it into a normalized database without losing or duplicating anything, **query** it, and **serve** it to people through a dashboard. You did the cleaning in Assignment 3 and the schema in Assignment 4. Now you write the code that fills that schema and the app that shows it.

Nothing here is copy and paste. The Guide notebook teaches every technique on a small practice dataset (a coffee shop's orders, in its own `practice` schema). You then write your own loader, your own SQL and your own dashboard for `EmployeeDB`. The term project's Phase 2 needs exactly these skills, so the time you spend here pays off twice.

What to do

1. **Work through `Assignment05_Guide.ipynb` first.** Run every cell and do the two short "Try it" exercises. Not graded, not submitted. Plan on about two hours.
2. **Copy `Assignment05_Starter.ipynb` to `A5_LastName_FirstName.ipynb`** and put your `Cleaned_data.csv` in the same folder.
3. **Part A:** write and run your loader, run it a second time, and reconcile the counts.
4. **Part B:** answer the 10 SQL questions in the starter notebook.
5. **Part C:** write `app.py` from the specification below, run it, take the screenshots and answer the 3 dashboard questions.
6. **Part D:** answer the 3 industry scenarios at the end of the starter notebook.
7. **Restart and run all** in your notebook, then submit the 3 files listed under Deliverables.

Part A: Load (35 points)

Where each table column comes from

These are the column names in a correct Assignment 3 file. If yours differ, rename them in pandas before loading. Columns not listed (for example `DivisionAddress`, `PersonalAddress`) are not loaded.

Table	Table column ← CSV column
`departments`	`departmentname` ← Department, `departmentdescription` ← DepartmentDescription
`jobtitles`	`jobtitle` ← JobTitle
`divisions`	`divtype` ← Division
`divisionlocations`	`divisionid` (looked up from Division), `divstreet` ← DivStreetAddress, `divcity` ← DivCity, `divstate` ← DivState, `divzip` ← DivZip, `divisionphone` ← DivisionPhone. One row per division.
`jobdescriptions`	`jobtitleid` and `divisionid` (looked up), `description` ← JobDescription. One row per job title and division pair.
`personaldetails`	`ssn` ← SocialSecurity, `name` ← Name, `honorific` (not in the CSV, leave NULL), `highestdegree` ← HighestDegree, `email` ← Email, `phonenumber` ← PhoneNumber, `birthdate` ← Birthdate, `perstreet` ← PerStreetAddress, `percity` ← PerCity, `perstate` ← PerState, `perzip` ← PerZip
`employeedetails`	`employeeid` ← EmployeeID, `ssn` ← SocialSecurity, `departmentid`, `jobtitleid`, `divisionid` (all looked up), `salary` ← Salary. One row per employee.

Loader rules (graded)

- **Clean in pandas before inserting.** `Birthdate` becomes a real date, not text. Print how many birthdates could not be read.
- **Reject, do not invent.** A row that cannot satisfy a NOT NULL or CHECK constraint goes into `rejects_employees.csv` and is counted. Never write a 0 or a made-up value to get past a constraint.
- **All or nothing.** The whole load runs in one transaction.
- **Safe to re-run.** Run your loader twice. The second run must add zero rows to every table.
- **Prove it.** One reconciliation table: for each of the 7 tables, the count expected from the CSV, the count after run 1, and the count after run 2.
- **Load report.** 4 to 6 sentences in the notebook: rows read, loaded and rejected (and why), which pattern you used for the three tables without a UNIQUE column, and one real problem you hit, with the error or wrong count you saw and how you fixed it.

Your schema is already built. Do not change it, except that you may *add* a constraint if your load report explains why.

Part B: SQL (20 points)

The 10 questions are in the starter notebook, written as business questions with no SQL given. Write each query yourself and leave its result showing. Every query must:

- filter by **names** (`'Engineering'`, `'Austin'`), never by an id number you looked up;

- let the database compute every average or count, never a number you typed in;
- show names, not ids, wherever a person would read the result.

Q10 also has a short written part about data quality.

Part C: Dashboard (35 points)

Write `app.py`. It must read from `EmployeeDB` (not the CSV) and contain:

#	Element	Specification
1	Sidebar filters	**Division** multiselect (all selected by default), **Department** selectbox with an "All" option, **Salary range** slider from the lowest to the highest salary in the database. **Every** number, chart and table below follows all three filters.
2	KPI row	Headcount, average salary and median salary for the current filters (`st.metric`).
3	Headcount by department	Bar chart.
4	Salary distribution by division	Box plot, one box per division.
5	Average salary by age band	Bar chart with bands Under 30, 30-39, 40-49, 50-59, 60+, shown in that order, with the number of employees in each band visible (label or hover). Compute age from `birthdate` in SQL.
6	Top 10 job titles by average salary	Horizontal bar chart, highest at the top.
7	Employee table	Inside an `st.expander`: name, job title, department, division and salary for the current filters, highest salary first.
8	Engineering	Database engine cached with `st.cache_resource`, query results with `st.cache_data`. Filter values reach SQL as parameters, not pasted into the SQL string. Every chart has a title and readable axis labels.

Plotly, matplotlib or seaborn are all fine. Layout and colors are your choice.

Dashboard questions (answer in the PDF)

1. Select only **West Division** (Department "All", full salary range). Report the three KPIs. Why are the average and the median salary so far apart? (1 to 2 sentences)
2. With no filters, the salary box plot is hard to read. Explain what causes this and use your filters to make the boxes readable. Include a screenshot of the readable version and say which filter settings you used.
3. Which age band has the highest average salary for the whole company? Now select Department **Operations**. Does the same band still lead, and how many employees is that average based on? What does this tell you about trusting an average? (2 to 3 sentences)

Part D: Industry scenarios (10 points)

The last section of the starter notebook describes three incidents of the kind data engineers handle at large companies. Answer each one in its own cell, in 3 to 5 sentences. There is no single right answer: refer to **your own** loader, SQL or `app.py` wherever you can, because a specific answer (the line, the query, the setting) beats a general one.

- **D1. The job said SUCCESS:** the HR system changes how it sends salaries and a large share of rows quietly stop loading.
- **D2. Two numbers for the same thing:** Finance's report and your dashboard disagree on an average salary.
- **D3. From 14,000 rows to 50 million:** time your loader, then explain what breaks first at a much larger size and what you would change.

Deliverables (submit all three)

#	Deliverable	What is in it
1	`A5_LastName_FirstName.ipynb`	Parts A, B and D, run top to bottom with every output showing (Kernel, Restart and Run All, then save).
2	`app.py`	Your dashboard. Replace your password with `"YOUR_PASSWORD"` before you submit.
3	`A5_LastName_FirstName.pdf`	Screenshots, each numbered with a one-line caption: (1) the terminal running `streamlit run app.py`; (2) the full dashboard with no filters; (3) the dashboard with West Division selected; (4) the readable box plot from question 2. Then your answers to the 3 dashboard questions.

Do not submit the Guide notebook, the practice files, your CSV or `rejects_employees.csv`.

If your Assignment 3 file cannot be loaded at all, a rescue `Cleaned_data.csv` will be posted on Canvas once Assignment 3 grades are released. Say in your load report which file you used.

Grading rubric (100 points)

Criterion	Excellent (full points)	Adequate (partial)	Insufficient (low/zero)	Points
A. Loader works	All 7 tables loaded from your CSV in the right order with correct foreign keys and the column map above; one transaction	Loads with a table missing or a foreign key wrong	Load fails, tables empty, or a provided or practice loader handed in unchanged	/15
A. Data rules	Birthdate stored as a date; unreadable count printed; bad rows rejected to a file and counted, nothing invented	One of these missing	Birthdate as text, or values invented to pass constraints	/8
A. Re-run and reconciliation	Loader run twice, all 7 counts identical and equal to the expected counts, shown in one table	Counts shown but not compared, or a table duplicates on the second run	No second run or no counts	/7
A. Load report	Numbers, patterns and one specific real problem with its fix	Present but vague ("I had some errors")	Missing	/5
B. SQL (10 × 2)	Each query correct, follows the three rules, result showing; Q10 written part names a concrete fix	1 point for a query with a wrong result, a hard-coded id or number, or ids where names belong	Missing, or result not showing	/20
C. Dashboard complete	`app.py` runs against `EmployeeDB`; elements 2 to 7 all present and correct	Runs with 1 or 2 elements missing or wrong	Does not run, or reads the CSV	/15
C. Filters	All three filters present and every KPI, chart and table follows them	Filters present but some charts ignore them	No working filters	/8
C. Engineering	Engine and query caching, parameterized filters, titles and labels	One of these missing	None of them	/5
C. PDF	4 captioned screenshots from your own machine; 3 answers correct and reasoned from your own dashboard	Screenshots or answers incomplete	Missing	/7

D1. The job said SUCCESS	Explains what your own loader does with the new salary format, why nothing failed, and a concrete check that would stop the run or alert someone	General answer not tied to your loader, or no concrete check	Missing	/4
D2. Two numbers	Three legitimate reasons (at least one from your own `app.py`) and a sensible way to settle it	Fewer than three reasons, or none from your app	Missing	/3
D3. 50 million rows	Your loader's time reported; names the specific part of your code that slows down, why, and a fix	No time reported, or a general answer not tied to your code	Missing	/3

Common deductions, read these BEFORE you submit

- **−8:** `birthdate` loaded as text or left NULL for everyone. The age questions and the age band chart depend on it.
- **−5:** a salary of 0 (or any made-up value) written to get past a constraint.
- **−5:** the second loader run duplicates rows.
- **−3:** `conn.close()` in the middle of `app.py`, so later charts fail.
- **−2:** notebook not run top to bottom, or outputs missing.
- **−2:** your real password left in `app.py`.
- **No penalty:** numbers that differ from a classmate's because your Assignment 3 file differs, or age results that shift by a day or two. Grading checks your query logic against your own data.

Stuck?

Ask in class first, and bring the exact error text. After that, the Discussion Board. On the board, post what you ran, the full error, and what you already tried; we answer within a day.